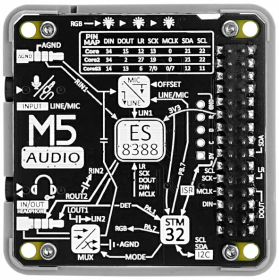
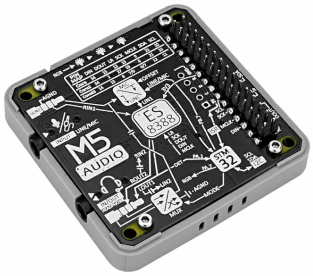
Module Audio

SKU:M144







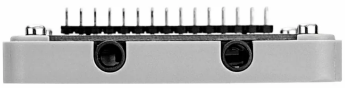


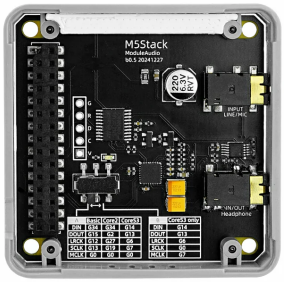


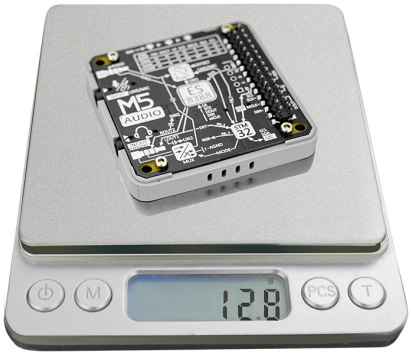












Description

Module Audio is an audio-interaction expansion module for M5Stack, built on the **ES8388** audio codec solution. It offers dual 3.5 mm jacks (**TRS** audio jack supports only input, while **TRRS** combo jack supports both microphone input and headphone playback.) to fulfill various recording and stereo playback needs. An onboard STM32G030F6P6 microcontroller handles TRRS jack insertion detection and drives WS2812C **RGB** LEDs. Register configuration enables automatic switching between CTIA (American) and OMTP (International) wiring standards, ensuring compatibility with most headsets featuring integrated microphones. This module is ideal for smart voice, interactive art, educational entertainment, portable recording, and other audio applications.

Features

- STM32G030F6P6 MCU control
- High-fidelity audio codec
- Dual-channel microphone input
- TRS + TRRS dual 3.5 mm jacks
- CTIA/OMTP wiring standard switching
- WS2812C RGB status LEDs
- Supported development platforms
  - UiFlow1
  - UiFlow2
  - Arduino IDE

Includes

- 1 × Module Audio

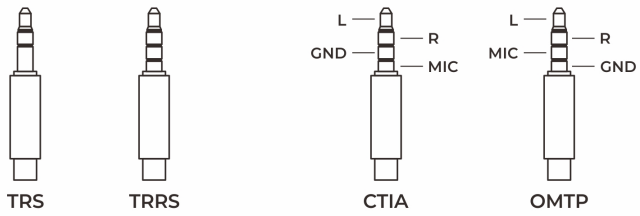
Applications

- Smart speakers
- Voice recognition
- Educational entertainment
- Interactive art installations

Specifications

Specification	Parameter
MCU	STM32G030F6P6, I2C Communication @0x33
Audio Codec	ES8388, I2C Communication @0x10
Audio I/O	2-channel mic input, 1-channel stereo headphone output
Jack Types	1 × TRS (mic-only), 1 × TRRS (mic + headphone)
Status LEDs	3 × WS2812C_2020
Headset Standard	CTIA / OMTP
Operating Current	DC 3.3 V @ 23.53 mA
Standby Current	DC 3.3 V @ 8.58 mA
Operating Temperature	0 ~ 40 °C
Product Size	54.0 × 54.0 × 13.1 mm
Product Weight	12.8 g
Package Size	132.0 × 95.0 × 16.0 mm
Gross Weight	26.3 g

Learn



Headphone Jack Types & Wiring Standards

In audio interfaces, common plug formats include TRS and TRRS:

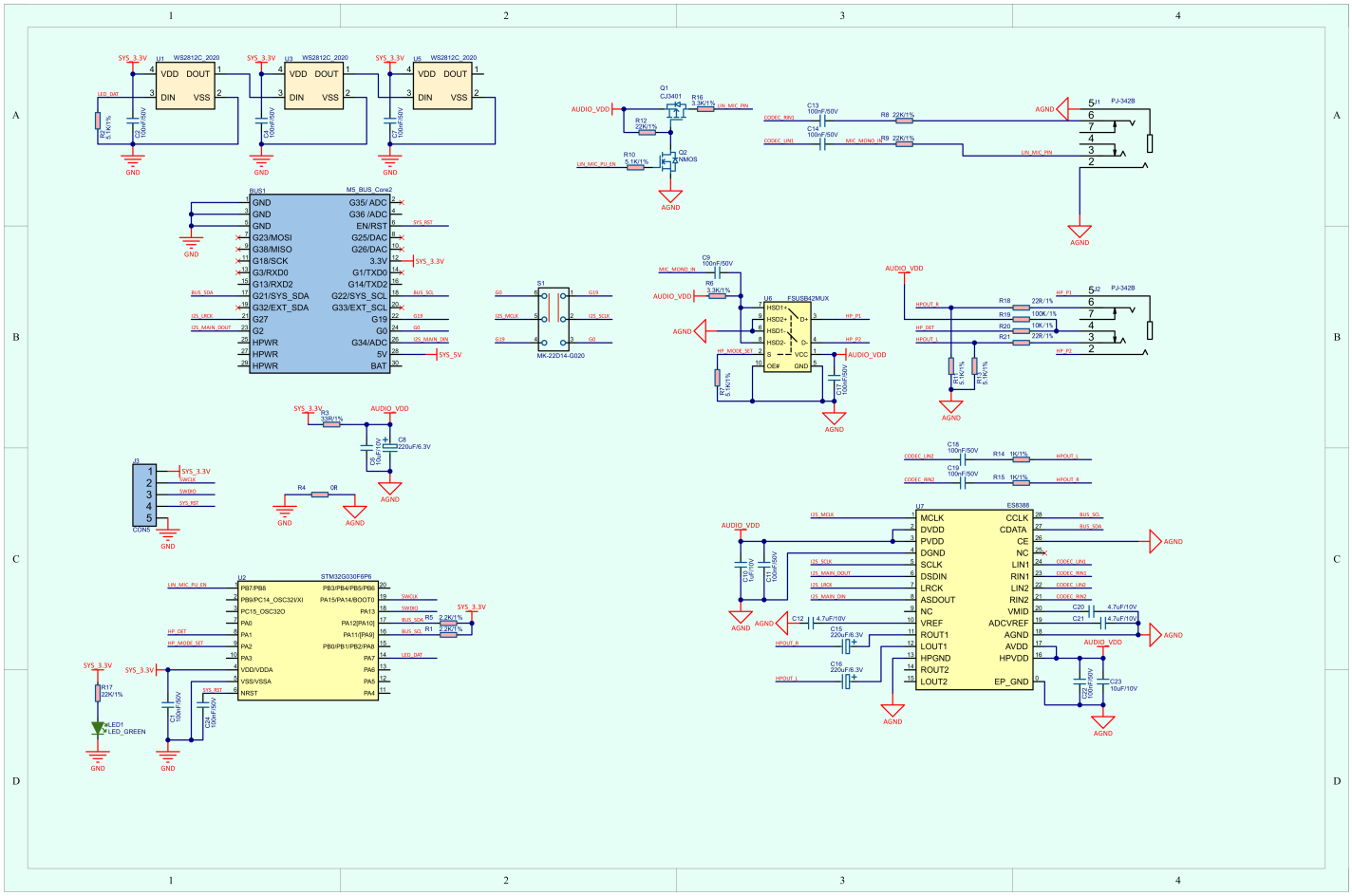
- **TRS (Tip-Ring-Sleeve)** three-contact plugs support only mono microphone input or stereo output; on this module it is mic-only;
- **TRRS (Tip-Ring-Ring-Sleeve)** four-contact plugs provide left/right stereo output plus a microphone channel for combined mic input and playback.

For TRRS plugs there are two wiring standards:

- **CTIA (American)**: L (Tip) = Left, R (Ring1) = Right, MIC (Ring2) = Microphone, GND (Sleeve) = Ground;
- **OMTP (International)**: L (Tip) = Left, R (Ring1) = Right, GND (Ring2) = Ground, MIC (Sleeve) = Microphone.

Schematics

- Module Audio Schematics PDF



PinMap

RGB & FSUSB42MUX & I2C

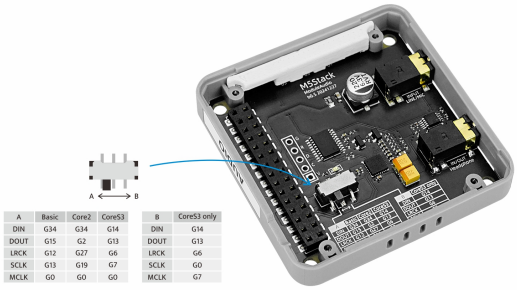
STM32G030F6P6	PA7	PA2	PA1	PA12	PA11
WS2812C	LED_DAT				
FSUSB42MUX		HP_MODE_SET	HP_DET		
I2C				SDA	SCL

Pin Details

Module Audio reserves A/B Pin Configuration for I2S pin switching to match host defaults:

- Configuration A matches Basic/Core2 default I2S pins
- Configuration B matches CoreS3 default I2S pins, since CoreS3’ s onboard ES7210 occupies the original I2S pins and requires

switching to avoid conflict



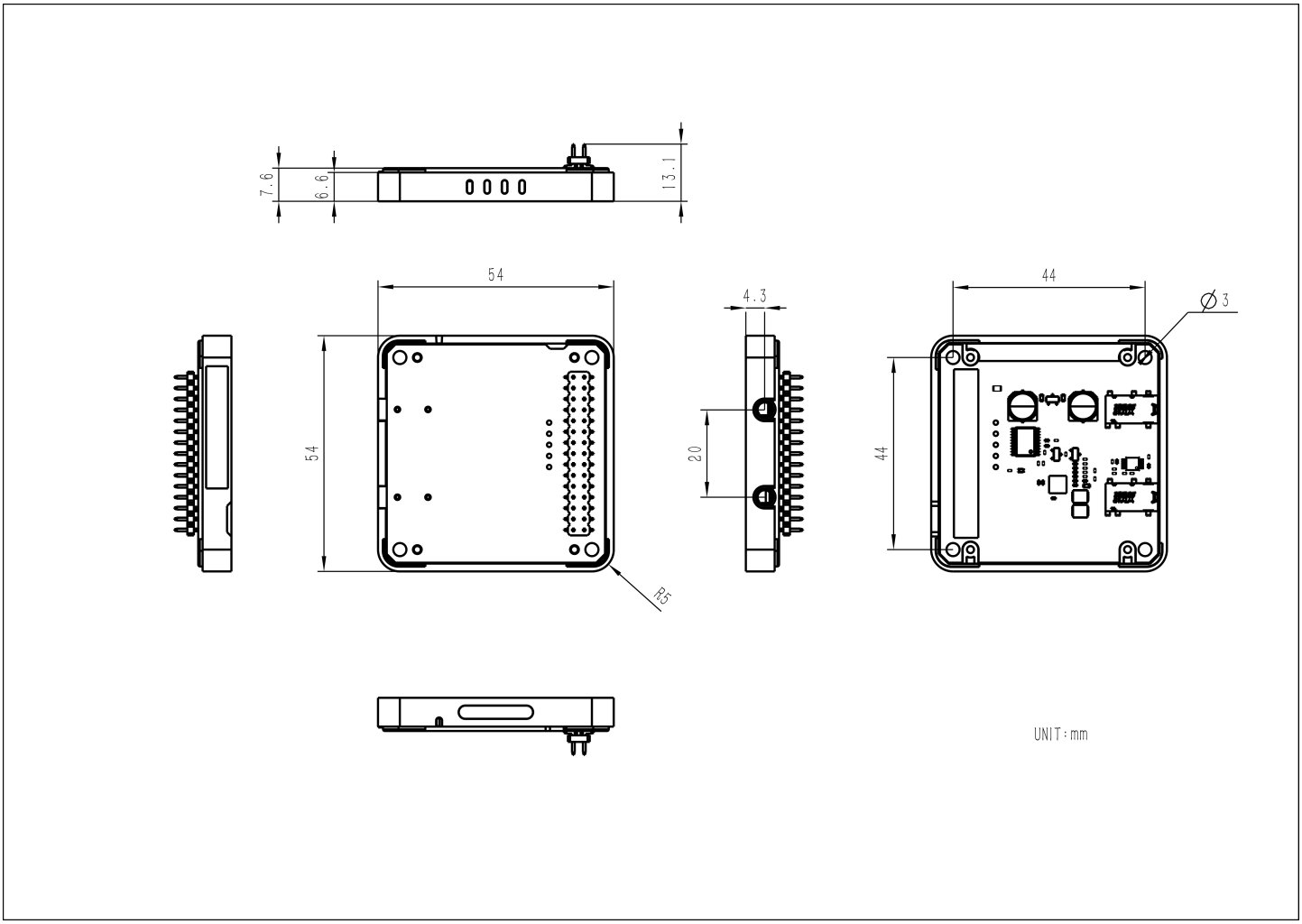
A	Basic	Core2	CoreS3
DIN	G34	G34	G14
DOUT	G15	G2	G13
LRCK	G12	G27	G6
SCLK	G13	G19	G7
MCLK	G0	G0	G0

B	CoreS3 only
DIN	G14
DOUT	G13
LRCK	G6
SCLK	G0
MCLK	G7

M5-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	
GND	3	4	
GND	5	6	RST
	7	8	
	9	10	
	11	12	3V3
	13	14	
	15	16	
SDA	17	18	SCL
	19	20	
I2S_LRCK	21	22	I2S_MCLK/I2S_SCLK
I2S_OUT	23	24	I2S_MCLK/I2S_SCLK
	25	26	I2S_IN
	27	28	5V
	29	30	

Model Size



Datasheets

- ES8388

Softwares

Arduino

- Module Audio Arduino Quick Start
- Module Audio Arduino Library

UiFlow1

- Coming Soon...

UiFlow2

- Module Audio UiFlow2 Docs

Protocol

- Module Audio I2C Protocol PDF

M5Stack Module-Audio I2C Protocol																	V1 (FW Version) 2026/8/11
REG MAP (Addr: 0x33)		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	note
Microphone config	0x00 R/W	Config																Config: Configure LINE/MIC on/off 0: Close 1: Open default: 1
Headphone Mode	0x10 R/W	Mode																Mode: Headphone mode selection 0: American Standard (CTIA) 1: National Standard (OMTP) default: 0
Headphone Insert Status	0x20 R	Status																Status: DET Status 0: Not insert 1: Insert
RGB Brightness	0x30 R/W	Brightness																Brightness: RGB brightness value: 0~100 default: 10
RGB LED	0x40 R/W	LED1_R	LED1_G	LED1_B	LED2_R	LED2_G	LED2_B	LED3_R	LED3_G	LED3_B								LED R/G/B: RGB value value: 0~255 default: 0
Save configuration to flash	0xF0 W	Write value																Write 1 to save to internal Flash
Firmware Version	0xF0 R															Version		Version: Software version number
I2C Address	0xF0 R/W																Address	Address: Device address value: 0x08~0x77 default: 0x33

Register Description
1. Microphone config: Configure the LINE/MIC to be enabled or disabled.
2. Headphone Mode: Configures the headphone mode. The default is national standard mode, with customizable settings.
3. Headphone Insert Status: 0 indicates device inserted, 1 indicates device not inserted.
4. RGB Brightness: Configures the brightness of the RGB lights. The default brightness is 10, and it can be customized.
5. RGB LED: Configures the color of the RGB lights. The default state is off.
6. Save configuration to flash: Write 1 to this register to save the current configuration to the internal Flash.
Configuration content includes:
Microphone config
Headphone Mode
RGB Brightness
7. I2C Address: Configures the I2C address. After successful setting, the data will be saved in internal Flash memory, will not be lost during power down, and will take effect immediately.

Notes
1. Writing to Flash requires first erasing the internal Flash, which takes some time, approximately 20ms.
2. To extend the lifespan of the Flash, avoid frequent write operations.
3. If the value to be written is the same as the current value stored in Flash, the write operation will be skipped to avoid unnecessary erase cycles, further extending the Flash' s lifespan.

Internal Firmware

- Module Audio Internal Firmware
- I2C Address

Chip	I2C Address
STM32G030F6P6	0x33
ES8388	0x10

Video

- Module Audio product introduction and demonstration

M144_Video.mp4